

AEGIS 27-Agent Architecture Guide

Detailed Breakdown of Operations, Roles, Input/Output Pipelines, and LLM Provider Routing

1. Core Engineering Suite

Architect Agent	Role: Software & System Architect Provider: NVIDIA NIM / Groq (Llama-3.1-70B)
How it Works:	Analyzes system requirements, devises microservices architecture, schema designs, data flows, and tech stack trade-offs.
Input Data:	Feature specs, user stories, scalability goals
Output Data:	Architecture diagrams (Mermaid), API contracts, data flow specs

Code Reviewer Agent	Role: Lead Code Reviewer & Security Auditor Provider: NVIDIA NIM / Groq (Llama-3.1-70B)
How it Works:	Audits PRs and codebases for bugs, anti-patterns, security flaws, performance bottlenecks, and adherence to style guides.
Input Data:	Source code diffs, code files, PR descriptions
Output Data:	Inline feedback, severity ratings (High/Med/Low), refactored code snippets

Coder Agent	Role: Full-Stack Software Engineer Provider: NVIDIA NIM / Groq (Llama-3.1-70B)
How it Works:	Implements features, writes frontend components, backend APIs, database migrations, and complex algorithms.
Input Data:	Technical requirements, function signatures, wireframes
Output Data:	Production-ready code implementation with error handling

Debugger Agent	Role: Root Cause Diagnostic Specialist Provider: NVIDIA NIM / Groq (Llama-3.1-70B)
How it Works:	Inspects stack traces, runtime logs, and exception dumps to pinpoint exact failure points and issue hotfixes.
Input Data:	Error logs, exception tracebacks, failing code
Output Data:	Root cause analysis, patch recommendation, bug fix

DevOps Lead Agent	Role: Infrastructure & CI/CD Specialist Provider: NVIDIA NIM / Groq (Llama-3.1-70B)
How it Works:	Generates Dockerfiles, Kubernetes manifests, GitHub Action workflows, Terraform scripts, and deployment pipelines.
Input Data:	Deployment target, environment requirements
Output Data:	CI/CD YAML files, containerization scripts, IAC manifests
Test Engineer Agent	Role: QA & Test Automation Specialist Provider: Groq (Llama-3.1-8B-Instant)
How it Works:	Generates unit tests, integration tests, end-to-end test scripts, edge-case assertions, and mock datasets.
Input Data:	Source code files, API endpoint schemas
Output Data:	PyTest / Jest test files, edge case matrices
Ops Agent	Role: Site Reliability & Systems Ops Specialist Provider: Groq (Llama-3.1-8B-Instant)
How it Works:	Monitors system health metrics, configures Prometheus/Grafana alerts, and creates incident response runbooks.
Input Data:	Server metrics, error rates, CPU/Memory logs
Output Data:	Incident reports, alert configurations, runbooks
LangChain Code Reviewer Agent	Role: LLM & Agent Pipeline Auditor Provider: NVIDIA NIM / Groq (Llama-3.1-70B)
How it Works:	Audits multi-agent pipelines, prompt templates, RAG vector retrieval, and token usage for efficiency and memory leaks.
Input Data:	LangChain / LlamaIndex Python code
Output Data:	Prompt optimizations, token reduction advice, pipeline refactoring
Hello World Agent	Role: Instant Sanity & Latency Tester Provider: Groq (Llama-3.1-8B-Instant)
How it Works:	Executes ultra-fast sub-second ping tasks to test system latency, provider health, and basic API connectivity.
Input Data:	Simple ping query or heartbeat test
Output Data:	Sub-second status JSON, latency metrics

2. Planning & Management Suite

Orchestrator Agent	Role: Central Task Classifier & Router Provider: NVIDIA NIM / Groq (Llama-3.1-70B)
How it Works:	Analyzes raw user prompts, classifies task intent, dynamically routes execution to the best agent, and formats final output.
Input Data:	Unstructured user prompts or multi-step tasks
Output Data:	Target agent selection, execution plan, synthesized response

Planner Agent	Role: Agile Project Planner & Roadmap Strategist Provider: NVIDIA NIM / Groq (Llama-3.1-70B)
How it Works:	Deconstructs large project goals into sprints, epics, user stories, task dependencies, and milestone schedules.
Input Data:	Project PRDs, high-level business goals
Output Data:	Sprint breakdowns, Gantt chart structures, Kanban tasks

Project / Meeting Assistant Agent	Role: Meeting Summarizer & Action Item Tracker Provider: Groq (Llama-3.1-8B-Instant)
How it Works:	Parses meeting transcripts, notes, and discussions to extract decisions, assignees, deadlines, and executive summaries.
Input Data:	Raw meeting transcripts, audio notes
Output Data:	Bullet-point meeting summaries, action items with owners

3. Business & Support Suite

Customer Support Agent	Role: Real-Time Support Representative Provider: Groq (Llama-3.1-8B-Instant)
How it Works:	Provides immediate, empathetic, customer-facing resolution for product inquiries, billing questions, and troubleshooting.
Input Data:	User support tickets, chat inquiries
Output Data:	Empathetic resolution messages, escalation flags

Sales Assistant Agent	Role: B2B Sales & Lead Qualification Specialist Provider: Groq / HuggingFace (Mixtral-8x7B)
How it Works:	Drafts cold outreach emails, qualifies incoming leads, generates sales pitch decks, and crafts objection-handling scripts.
Input Data:	Lead profiles, product value propositions
Output Data:	Personalized email sequences, lead qualification scores

HR / Recruiter Agent	Role: Talent Acquisition & HR Policy Specialist Provider: Groq / HuggingFace (Mixtral-8x7B)
How it Works:	Generates job descriptions, screens candidate resumes against role criteria, and answers employee HR policy queries.
Input Data:	Resumes, job requirement specifications
Output Data:	Candidate match reports, interview questions, job posts

Legal Agent	Role: Corporate & Compliance Legal Specialist Provider: NVIDIA NIM / Groq (Llama-3.1-70B)
How it Works:	Reviews contracts, terms of service, privacy policies, NDAs, and regulatory compliance guidelines for legal risks.
Input Data:	Contracts, terms documents, regulatory text
Output Data:	Risk assessments, clause redlines, compliance checklists

Marketing Agent	Role: Growth & Campaign Marketing Strategist Provider: Groq / HuggingFace (Mixtral-8x7B)
How it Works:	Creates ad copy, social media campaigns, SEO strategies, content calendars, and brand messaging strategies.
Input Data:	Target demographics, campaign objectives
Output Data:	Ad copies, content calendars, SEO keyword maps

Finance Agent	Role: Financial Analyst & Budget Planner Provider: NVIDIA NIM / Groq (Llama-3.1-70B)
How it Works:	Analyzes balance sheets, cash flow statements, project budgets, unit economics, and detects spending anomalies.
Input Data:	Financial spreadsheets, revenue data
Output Data:	Forecast tables, anomaly alerts, budget variance reports

4. Specialized & Utility Suite

Doc Writer Agent	Role: Technical Documentation Specialist Provider: NVIDIA NIM / Groq (Llama-3.1-70B)
How it Works:	Produces clean, comprehensive technical documentation, API reference guides, Swagger specs, and README files.
Input Data:	Code repositories, API schemas
Output Data:	Markdown documentation files, OpenAPI specs

Writer Agent	Role: Creative Content & Copywriter Provider: Groq / HuggingFace (Mixtral-8x7B)
How it Works:	Drafts long-form blog posts, whitepapers, thought leadership articles, newsletters, and engaging stories.
Input Data:	Topic outlines, target audience guidelines
Output Data:	Polished articles, newsletters, stories

Email Assistant Agent	Role: Executive Communication Specialist Provider: Groq (Llama-3.1-8B-Instant)
How it Works:	Drafts, refines, and formats professional emails, executive announcements, follow-ups, and inbox replies.
Input Data:	Rough notes, email thread history
Output Data:	Polished email drafts with subject lines

Translator Agent	Role: Polyglot Localization Specialist Provider: Groq (Llama-3.1-8B-Instant)
How it Works:	Translates text across 50+ languages while preserving technical accuracy, cultural nuances, and context.
Input Data:	Source text, target language, context notes
Output Data:	Accurately localized translated text

Travel Planner Agent	Role: Itinerary & Travel Concierge Provider: Groq / HuggingFace
How it Works:	Builds customized travel itineraries, flight/hotel recommendations, budget breakdowns, and daily schedules.
Input Data:	Destination, duration, budget, preferences
Output Data:	Day-by-day itineraries, cost breakdowns

Health Tracker Agent	Role: Wellness & Fitness Coach Provider: Groq / HuggingFace
How it Works:	Tracks workouts, nutrition plans, calorie budgets, sleep metrics, and generates personalized fitness schedules.
Input Data:	Fitness goals, dietary restrictions, workout logs
Output Data:	Structured meal plans, exercise routines

Home Automation Agent	Role: Smart Home & IoT Controller Provider: Groq (Llama-3.1-8B-Instant)
How it Works:	Writes Home Assistant YAML configurations, Zigbee routines, Node-RED flows, and IoT automation scripts.
Input Data:	Smart devices list, desired automation triggers
Output Data:	Home Assistant YAML, automation logic

5. Skill-Creator Suite

Analyzer Agent	Role: Deep Pattern & Skill Data Deconstructor Provider: NVIDIA NIM / Groq (Llama-3.1-70B)
How it Works:	Deconstructs dataset inputs, prompt patterns, and workflows to extract core skills and reusable templates.
Input Data:	Raw execution logs, prompt datasets
Output Data:	Skill blueprints, prompt templates, pattern maps

Comparator Agent	Role: Side-by-Side Evaluation Specialist Provider: NVIDIA NIM / Groq (Llama-3.1-70B)
How it Works:	Compares multi-model responses, alternative code solutions, or prompt versions to identify pros, cons, and diffs.
Input Data:	Two or more solution candidates / responses
Output Data:	Comparison matrix, differential analysis

Grader Agent	Role: Automated Quality Assessment & Scoring Provider: NVIDIA NIM / Groq (Llama-3.1-70B)
How it Works:	Evaluates agent outputs against strict rubric criteria, assigning numerical quality scores and improvement suggestions.
Input Data:	Agent response output, target evaluation rubric
Output Data:	Quality scores (0-100), rubric breakdown, fix suggestions